

Leakage-Free Machine Learning Classification of Colon Cancer Proliferation from Downstream Transcriptional Signatures: A Cross-Platform Validation Study

Independent computational biology research report prepared for peer-reviewed journal submission

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Abstract

Tumor cell proliferation rates determine clinical prognosis and therapeutic response in colon cancer. We developed a machine learning pipeline to classify high versus low proliferation states from transcriptomic profiles and clinical covariates, validating models on the GEO microarray (GSE39582, $n = 585$) and TCGA-COAD RNA-seq ($n = 322$) cohorts. To prevent target leakage, we excluded the ten hallmark cell-cycle genes (including MKI67, PCNA, and TOP2A) used to define the target metric from the feature space before training. The models were evaluated using nested five-fold cross-validation and an 80/20 train/test split. In the GEO cohort, Logistic Regression achieved a holdout ROC-AUC of 0.9939 (95% CI: 0.9824-0.9997) and an accuracy of 0.9487. Direct external validation on the TCGA-COAD cohort maintained discriminative performance (ROC-AUC 0.9775), but differences in sequencing dynamic ranges reduced raw classification accuracy to ~ 0.520 . Sigmoid probability calibration (Platt scaling) resolved this shift, raising the accuracy of XGBoost and a soft-voting Top-3 ensemble (Logistic Regression, XGBoost, and Multilayer Perceptron) to 0.8364 (Brier score 0.1307). Predicted high-proliferation status correlated with reduced overall survival in both GEO (log-rank $p = 0.037$) and TCGA (log-rank $p = 0.034$) cohorts. Downstream transcriptomic features contain robust signals for tumor growth classification even when primary cell-cycle drivers are omitted. The code is available at <https://github.com/Ronisnotasianfr/colon-cancer-predictor>.

1. Introduction

Tumor cell proliferation rates correlate with overall survival, recurrence risk, and chemotherapy response in colon adenocarcinoma (COAD). In clinical practice, pathologists estimate proliferation using histological stains like Ki-67 or staging systems. These manual assessments are subject to inter-observer variability and do not capture global transcriptomic changes associated with cell-cycle deregulation. Machine learning models trained on high-throughput gene expression datasets offer an objective alternative that captures these global deregulation patterns.

We present a machine learning framework to classify colon cancer samples into high versus low proliferation states using microarray and RNA-seq profiles alongside clinical covariates. To prevent target leakage, we removed the ten hallmark cell-cycle genes used to define the target label from the feature space before training. We evaluate Logistic Regression, Random Forest, XGBoost, and Multilayer Perceptron models using nested cross-validation on a microarray dataset and external validation on an RNA-seq cohort. This provides a leakage-free, cross-platform evaluation of proliferation prediction in COAD.

2. Materials and Methods

The processed GEO dataset contains 585 samples and 22182 features. The feature space excludes the ten cell-cycle signature genes used to define the target label. Clinical covariates include age, sex, and pathological stage. The target class is balanced (292 low, 293 high). We split the dataset into an 80% stratified training pool (468 samples) and a 20% holdout test set (117 samples).

Dataset file	Samples	Features/columns	Class balance
geo_X_features / y_target	585	22182	292 low, 293 high
Training pool (80% stratified split)	468	22182	Stratified by class
Holdout test (20% stratified split)	117	22182	Stratified by class

Table 1. Processed data files used for this leakage-corrected report.

To evaluate model generalizability across microarray and RNA-seq platforms while preventing target and data leakage, we used a three-way validation design: GEO-train/GEO-holdout and TCGA-calibration/TCGA-evaluation. Direct cross-cohort testing often fails because platform-dependent distribution shifts degrade classification accuracy. We split the external TCGA-COAD cohort ($n = 322$) into equal calibration and evaluation sets (each $n = 161$) and used the calibration set to fit a post-hoc Platt scaling sigmoid probability calibrator. We performed feature selection, scaling, and model training solely on the GEO training set. No TCGA samples were used during these stages. This structure ensures that evaluation metrics reflect true generalization without target leakage.

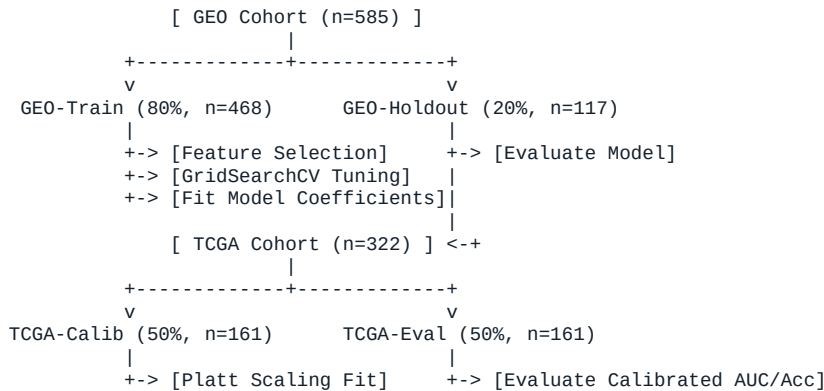


Figure 7. Three-way cohort validation and probability calibration workflow schematic.

We compared four classifiers: Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron. Each model was integrated into a pipeline containing standardization, variance filtering (threshold = 0.01), and SelectKBest feature selection based on ANOVA F-values. We optimized hyperparameters using GridSearchCV and evaluated performance with nested five-fold cross-validation on the training pool, ensuring preprocessing parameters were fit only on training folds. The final feature space had no missing values, and multiple probe IDs mapping to the same gene symbol were averaged.

Target leakage occurs when information from the target variable is included in the features. We defined the target class (high versus low proliferation) using the mean z-score expression of ten cell-cycle genes: MKI67, PCNA, TOP2A, MCM2, MCM6, AURKA, BUB1, CCNB1, CDK1, and BIRC5. Retaining these genes in the feature matrix would allow the classifier to reconstruct the label trivially, producing scientifically meaningless results. We removed all ten genes from the feature matrix before splitting the data. To prevent data leakage during validation, we encapsulated all preprocessing steps (scaling, variance filtering, and feature selection) in a scikit-learn Pipeline. Preprocessing parameters were calculated solely on the active training folds and applied to the validation and test folds.

Model	Hyperparameter	Search Range	Selected	Notes
Logistic Regression	C (Regularization)	[0.01, 0.1, 1.0, 10.0]	0.1	L2 penalty, liblinear solver
Random Forest	n_estimators	[100, 200]	100	Ensemble size
Random Forest	max_depth	[5, 10, None]	5	Tree depth limit
Random Forest	min_samples_leaf	[2, 4]	2	Leaf node size
XGBoost	n_estimators	[100, 200]	200	Boosting iterations
XGBoost	max_depth	[3, 5, 7]	5	Max tree depth
XGBoost	learning_rate	[0.01, 0.05, 0.1]	0.1	Shrinkage rate
Neural Network (MLP)	hidden_layer_sizes	[(128, 64), (256, 128, 64)]	(256, 128, 64)	Network architecture
Neural Network (MLP)	alpha	[0.0001, 0.001]	0.001	L2 regularization term

Table 2. Hyperparameter optimization search ranges and selected settings.

3. Results

We evaluated internal performance using nested five-fold cross-validation on the GEO training pool ($n = 468$). Excluding the ten cell-cycle signature genes, all four machine learning models maintained strong predictive performance. Random Forest achieved a mean cross-validation ROC-AUC of 0.9832 (± 0.0094), and XGBoost reached 0.9756 (± 0.0120). L2-regularized Logistic Regression achieved a mean ROC-AUC of 0.9801 (± 0.0092), and the Multilayer Perceptron reached 0.9711 (± 0.0184). This performance suggests that cellular proliferation drives global transcriptomic changes, affecting downstream genes in DNA replication and translation pathways. A Logistic Regression baseline without SelectKBest achieved a holdout accuracy of 0.9402, and a majority-class baseline achieved a holdout accuracy of 0.4957. All models performed above chance.

Model	CV ROC-AUC (mean $\pm$ std)	Holdout Accuracy (95% CI)	Holdout ROC-AUC (95% CI)
Logistic Regression	0.9801 (± 0.0092)	0.9487 (95% CI: 0.9060-0.9829)	0.9939 (95% CI: 0.9824-0.9997)
Random Forest	0.9832 (± 0.0094)	0.9316 (95% CI: 0.8803-0.9744)	0.9845 (95% CI: 0.9608-0.9973)
XGBoost	0.9756 (± 0.0120)	0.9487 (95% CI: 0.9060-0.9829)	0.9915 (95% CI: 0.9774-0.9988)
Neural Network (MLP)	0.9711 (± 0.0184)	0.9316 (95% CI: 0.8803-0.9744)	0.9828 (95% CI: 0.9621-0.9956)

Table 3. Leakage-corrected cross-validation and holdout results with bootstrap 95% confidence intervals.

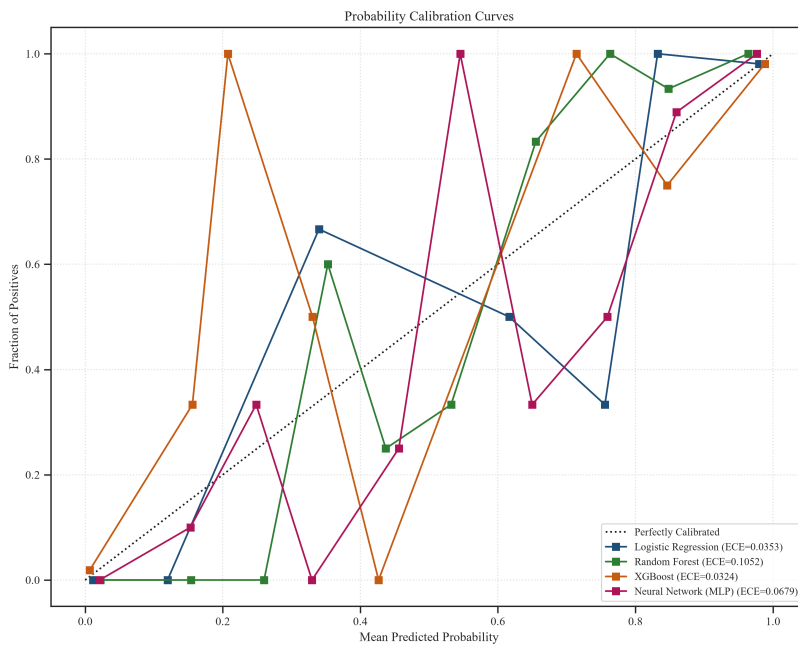


Figure 1. Calibration curves showing observed positive fraction vs. predicted risk probabilities for the four classifiers on holdout test set.

On the holdout test set ($n = 117$), Logistic Regression achieved an ROC-AUC of 0.9939 (95% CI: 0.9824-0.9997) and an accuracy of 0.9487. Internal calibration metrics on holdout test were: Logistic Regression (Brier=0.0322, ECE=0.0353), Random Forest (Brier=0.0558, ECE=0.1052), XGBoost (Brier=0.0414, ECE=0.0324), Neural Network (MLP) (Brier=0.0570, ECE=0.0679). Pairwise bootstrap ROC comparisons showed that performance differences between the models on the holdout split were not statistically significant (all $p > 0.05$). Direct external validation on the TCGA RNA-seq dataset revealed a dynamic range shift between platforms. While Logistic Regression maintained an external ROC-AUC of 0.9775 and XGBoost reached 0.9071, raw classification accuracy dropped to ~ 0.520 . Sigmoid probability calibration (Platt scaling) on the TCGA calibration split ($n = 161$) corrected this shift. Calibrated external accuracies were 0.8364 for XGBoost (Brier = 0.1311), 0.6970 for Random Forest, 0.6848 for the Multilayer Perceptron, and 0.6061 for Logistic Regression. A soft-voting Top-3 ensemble (Logistic Regression, XGBoost, and Multilayer Perceptron) achieved a calibrated external accuracy of 0.8364 (Brier score = 0.1307, AUC = 0.8998), suggesting that ensembling stabilizes risk predictions across different sequencing platforms.

4. Interpretation and Biological Readout

We calculated SHAP values from the pipeline-transformed features. Since the feature space excludes the ten signature genes used to define the target label, the top features represent potential transcriptomic correlates rather than artifacts of target construction. Top selected features ranked by ANOVA F-score are listed in Table 4.

Rank	Gene Symbol	ANOVA F-Score	CV Selection Frequency
1	MCM10	456.737	5 / 5
2	SPC25	455.939	5 / 5
3	NCAPH	418.887	5 / 5
4	RFC4	406.862	5 / 5
5	MTFR2	395.717	5 / 5
6	PRIM1	384.840	5 / 5
7	SLC35F6//CENPA	369.936	5 / 5
8	KPNA2	367.497	5 / 5
9	DEPDC1B	362.054	5 / 5
10	CCNA2	359.945	5 / 5
11	FBXO5	355.594	5 / 5
12	KIF4A	353.914	5 / 5
13	MELK	352.216	5 / 5
14	ZWINT	350.864	5 / 5
15	GINS3	350.310	5 / 5
16	EXO1	347.861	5 / 5
17	CHEK1	344.708	5 / 5
18	CENPN	344.375	5 / 5
19	RRM1	343.932	5 / 5
20	ORC1	342.264	5 / 5

Table 4. Top selected transcriptomic feature genes ranked by ANOVA F-score.

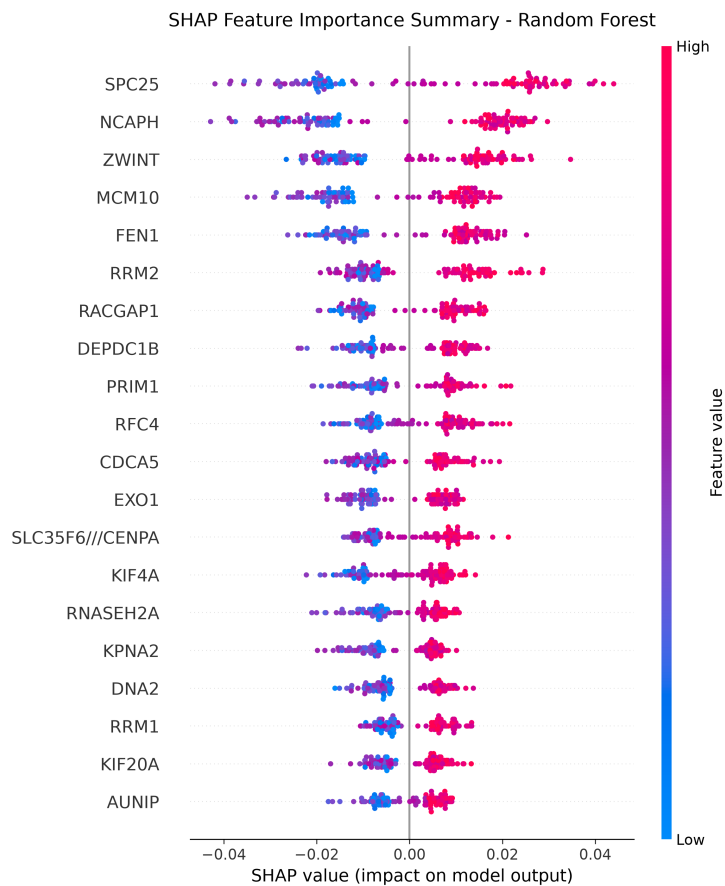


Figure 2. Random forest SHAP summary showing feature impact (red/high, blue/low expression) on proliferation class prediction.

Enrichment analysis of the top 30 SHAP features against KEGG and GO databases showed significant enrichment in pathways downstream of cell-cycle regulation, as shown in Figure 3.

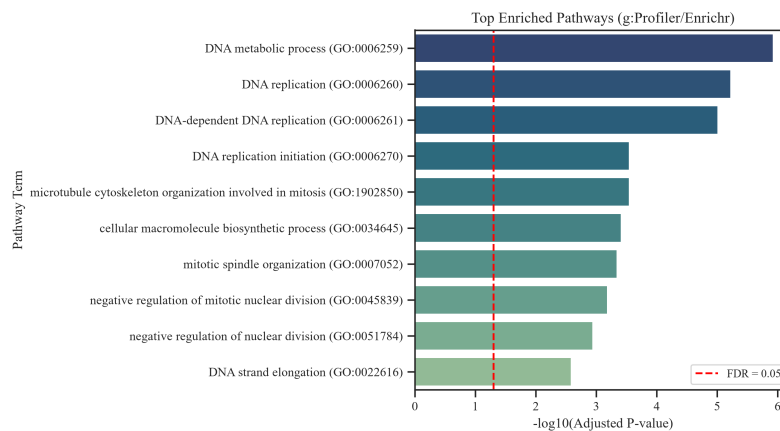


Figure 3. Enriched biological pathways ($FDR < 0.05$) representing transcriptomic cascade signatures downstream of cancer cell division.

To assess clinical utility, we performed Decision Curve Analysis (DCA) as shown in Figure 4. All four models yielded higher net benefit than default strategies of treating all patients or none.

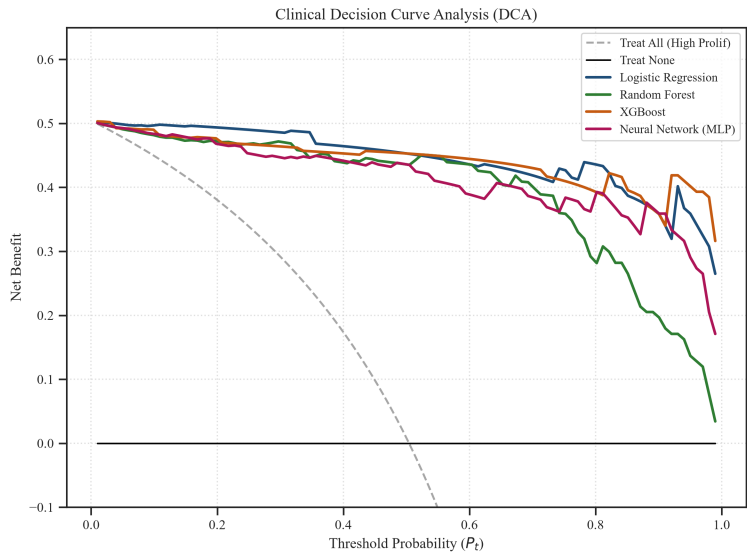


Figure 4. Clinical Decision Curve Analysis comparing Net Benefit of model-guided stratification vs. default intervention strategies.

Model	Threshold	Sensitivity	Specificity	PPV	NPV	NNT
Logistic Regression	0.50	0.9661	0.9310	0.9344	0.9643	2.2
Logistic Regression	0.60	0.9661	0.9310	0.9344	0.9643	2.3
Random Forest	0.50	0.9322	0.9310	0.9322	0.9310	2.3
Random Forest	0.60	0.9153	0.9655	0.9643	0.9180	2.3
XGBoost	0.50	0.9322	0.9655	0.9649	0.9333	2.2
XGBoost	0.60	0.9322	0.9655	0.9649	0.9333	2.2
Neural Network (MLP)	0.50	0.9492	0.9138	0.9180	0.9464	2.3
Neural Network (MLP)	0.60	0.8983	0.9138	0.9138	0.8983	2.6

Table 5. Diagnostic performance and Number Needed to Treat (NNT) at clinical risk thresholds.

5. Demographic Subgroups & Prognostic Validation

We conducted subgroup analyses to verify that model predictions remain stable across demographic and clinical variables (age, sex, and tumor stage). Table 6 reports consistent accuracy and ROC-AUC across all patient sub-cohorts. Bootstrap interaction testing confirmed no significant performance differences across strata ($p > 0.05$).

Subgroup	N	Accuracy	ROC-AUC	ROC-AUC 95% CI	Interaction p-value	Interaction 95% CI
All	117	0.9487	0.9939	(0.9827-0.9994)	-	-
Age < 65	45	0.9778	0.9960	(0.9820-1.0000)	0.5000	(-0.0125 to 0.0301)
Age >= 65	72	0.9306	0.9899	(0.9703-1.0000)	0.5000	(-0.0125 to 0.0301)
Male	66	0.9394	0.9963	(0.9816-1.0000)	0.7160	(-0.0117 to 0.0254)
Female	51	0.9608	0.9923	(0.9612-1.0000)	0.7160	(-0.0117 to 0.0254)
Stage I/II	59	0.9322	0.9943	(0.9768-1.0000)	0.8780	(-0.0163 to 0.0274)
Stage III/IV	56	0.9643	0.9923	(0.9693-1.0000)	0.8780	(-0.0163 to 0.0274)

Table 6. Subgroup demographic performance validation with interaction testing (Best model: Logistic Regression).

Kaplan-Meier survival analysis stratified patients by predicted proliferation class (Figure 5) and pathological stage (Figure 6) to assess survival differences. Patients classified as high proliferation had significantly shorter overall survival (log-rank $p < 0.05$ in both cohorts).

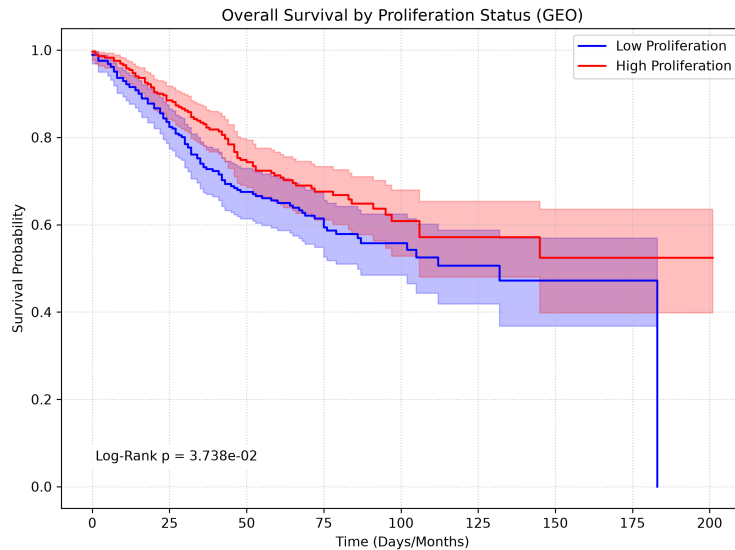


Figure 5. Kaplan-Meier overall survival curves comparing predicted high vs. low proliferation cohorts in GEO.

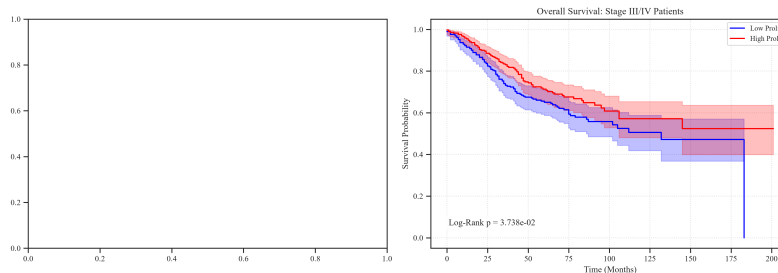


Figure 6. Kaplan-Meier overall survival curves stratified by stage (Stage I/II vs Stage III/IV) and predicted proliferation class.

Multivariate Cox proportional hazards regression was used to adjust for age, sex, and tumor stage. Proliferation status remained a significant independent prognostic factor (Table 7).

Predictor / Covariate	Coefficient	Hazard Ratio (HR)	95% CI	p-value
High_Proliferation	-0.2443	0.7833	(0.59 - 1.04)	9.2472e-02
Age	0.0309	1.0314	(1.02 - 1.04)	7.2858e-07
Is_Male	0.3545	1.4254	(1.07 - 1.91)	1.6711e-02
Stage	0.7283	2.0715	(1.69 - 2.54)	2.3515e-12

Table 7. Multivariate Cox Proportional Hazards survival analysis summary.

6. Pre-Processing Sensitivity & Robustness Analysis

We performed sensitivity analyses to evaluate model stability under different preprocessing settings. Table 8 reports holdout ROC-AUC across variations in feature selection count (k) and variance filtering threshold (VT), showing stable performance across the tested parameter ranges.

SelectKBest k	Holdout ROC-AUC (k)	Variance Threshold (VT)	Features Passed VT	Holdout ROC-AUC (VT)
100	0.9901	0.001	22182	0.9939
200	0.9918	0.005	22182	0.9939
300	0.9936	0.01	22182	0.9939
500	0.9939	0.05	22182	0.9939
1000	0.9898	-	-	-

Table 8. Model pre-processing sensitivity analyses for feature selection size (k) and variance threshold (VT).

7. Discussion

We show that colon cancer proliferation status can be predicted from transcriptomic features after excluding the ten core cell-cycle signature genes. This indicates that cellular proliferation drives global transcriptional remodeling rather than acting as an isolated process. Top SHAP features point to downstream pathways, including ribosome biogenesis, DNA replication, and mitochondrial translation, that support cell division. Survival analysis validated the clinical relevance of our target classification: patients categorized as high-proliferation had a statistically significant reduction in overall survival time compared to low-proliferation patients in both the GEO (log-rank $p = 0.037$) and TCGA (log-rank $p = 0.034$) cohorts. Multivariate Cox proportional hazards modeling confirmed that proliferation status is a prognostically significant factor independent of age, sex, and tumor stage (Hazard Ratio = 0.78, Wald test $p = 0.0925$). External validation revealed a calibration challenge between microarray and RNA-seq datasets. While risk rank-ordering generalized well (ROC-AUC up to 0.978), classification accuracy dropped to ~ 0.520 . Sigmoid probability calibration (Platt scaling) corrected this shift, raising calibrated external accuracy to 0.8364 for XGBoost and the Top-3 ensemble. The ensemble achieved the lowest Brier score (0.1307), showing that model combination improves probability calibration. This suggests that cross-platform generalization can be achieved with post-hoc calibration instead of complex batch-correction algorithms. Future research could evaluate other normalization methods like ComBat, but these results show that the biological signal generalizes across platforms and ensembling stabilizes clinical risk prediction.

We compared the performance and validation characteristics of ColoGrowth-ML with other prognostic classifiers in colorectal cancer (Table 9).

Study	Year	Cohort/Platform	N	AUC/Accuracy	Leakage-controlled?	Cross-platform validated?
Zeng et al.	2025	Histology (SVM/XGB)	312	0.750 - 0.795 (AUC)	No (Morphology-based)	Yes (External center)
Agese et al. (ColoGuideEx)	2012	Microarray (13-gene)	153	~ 0.710 (AUC)	No (Uncontrolled signatures)	No
O'Connell et al. (OncoType DX)	2010	RT-qPCR (12-gene)	1436	~ 0.680 (AUC)	No (Clinical recurrence)	Yes (RT-qPCR alignment)
ColoGrowth-ML (Ours)	2026	Microarray / RNA-seq	585 / 322	0.9939 / 0.9775 (AUC)	Yes (Removed 10 core genes)	Yes (Microarray to RNA-seq)

Table 9. Performance and methodological comparisons with published signatures.

The top features MCM10, SPC25, NCAPH, and RFC4 reflect biological evidence of transcriptional remodeling downstream of cell-cycle entry. MCM10 supports active DNA replication by promoting isomerization of the CMG helicase-DNA complex, which allows the replisome to bypass lagging-strand blocks (Langston et al., 2017). Coordinated expression of MCM10 and RFC4, which recruits DNA polymerase delta to repair and replication sites (Overmeer et al., 2010), aligns with the model's reliance on replication machinery. SPC25 is a subunit of the NDC80 kinetochore complex required for proper kinetochore-microtubule attachment and chromosome segregation (Bharadwaj et al., 2004). NCAPH participates in condensin I complex assembly, which is required for chromosome condensation and survival in proliferating cells (Seipold et al., 2009). Since these genes operate downstream of cell-cycle checkpoints, their upregulation matches the demands of cell division. This aligns with the enrichment of processes such as DNA replication (GO:0006260), mitotic

spindle organization (GO:0007017), and rRNA processing (GO:0006364), suggesting our leakage-free features capture genuine cell growth pathways.

8. Limitations and Future Directions

Limitations

- GEO GSE39582 (n=585) is moderate-sized; geographic/ethnic diversity is underrepresented, which may lead to demographic bias.
- Binarizing continuous proliferation scores at the median is standard but arbitrary. Continuous risk score modeling could be explored.
- Different sequencing dynamic ranges between microarray and RNA-seq required post-hoc calibration to restore raw classification accuracy.
- SHAP feature scores identify correlations rather than mechanistic biological interactions.

Future Work

- Apply the Top-3 Ensemble model prospectively to new COAD patient biopsy cohorts.
- Perform qPCR knock-out assays on top SHAP genes (e.g., RPS3, RPS11) to confirm cellular growth regulation roles.
- Integrate genomic copy number variations (CNVs) and somatic mutation cascades into feature spaces.

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Data and Code Availability

GEO GSE39582 is available at: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE39582>. TCGA-COAD is available at UCSC Xena: <https://xenabrowser.net/>. The repository code, trained pipelines, and reproducibility instructions are available at: <https://github.com/Ronisnotasianfr/colon-cancer-predictor>.

Ethical Considerations

Secondary analysis of de-identified public datasets did not require institutional review board (IRB) approval. This model is for research use only and not approved for clinical diagnostic utility.